

Numerical Methods in Astrophysics (0321488101): Numerov Method for Bound States in the One-Dimensional Schrödinger Equation

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1 Introduction

This project studies the one-dimensional time-independent Schrödinger equation using the Numerov method combined with parity-aware shooting procedures. The goals are to validate the method on systems with known analytic solutions, demonstrate numerical convergence and accuracy, and then apply it to more interesting bound-state potentials such as a double well and a finite square well. Extensions to scattering problems are also explored as a secondary component.

This problem is a boundary-value eigenvalue problem rather than a standard initial-value problem: only special energies produce wavefunctions satisfying the required boundary behavior. The Numerov method is well suited to this task because the time-independent Schrödinger equation has the second-order form $\psi'' = q(x)\psi$, with no first-derivative term. The project therefore focuses not only on obtaining eigenvalues, but also on checking that the numerical results are physically meaningful through analytic benchmarks, convergence tests, parity and node structure, and probability conservation in the scattering extension.

The full code was written in Python and is available in [this repository](#). Explanations of the code structure, usage, and implementation details are also provided there in the [README.md](#), the [code.walkthrough.md](#) guide, and comments throughout the source files. The implementation is modular, with separate components for the Numerov integrator, shooting method, root finding, analysis utilities, and an automated validation suite. This structure allows each numerical component to be validated independently and makes the overall workflow easier to understand.

2 Computational approach

2.1 The Schrödinger equation and Numerov method

In dimensionless units, the Schrödinger equation is written as:

$$\psi''(x) = 2[V(x) - E]\psi(x) \tag{1}$$

The wavefunction is integrated with the Numerov scheme on a uniform grid. For the symmetric bound-state potentials studied here, $V(-x) = V(x)$, so the eigenstates can be chosen to have definite parity: even states satisfy $\psi(-x) = \psi(x)$ and odd states satisfy $\psi(-x) = -\psi(x)$. That symmetry implies exact conditions at the origin, $\psi'(0) = 0$ for even states and $\psi(0) = 0$ for odd states, so the solver can integrate only on the half-domain $[0, x_{\max}]$ and reconstruct the other half by reflection. Eigenvalues are found by scanning for sign changes in a scalar mismatch function and then refining the resulting brackets with robust root finding; in the outward Numerov solver this is bisection followed by a few safeguarded secant-style polishing steps, while the inward-decay and RK4 solvers use straight bisection. The correctness of the numerical approach is verified through comparison with analytical solutions where available, through systematic convergence studies, and through direct automated tests of the implementation.

The overall numerical workflow is implemented in a modular way in the accompanying Python code. The core components include a Numerov integrator for propagating the wavefunction, a shooting module that enforces parity-based boundary conditions and constructs the mismatch function, and analysis utilities for convergence studies and parameter sweeps. This separation of concerns allows each numerical component to be validated independently and makes the implementation easier to interpret [1].

2.2 Shooting and root finding

The boundary-value eigenproblem is reformulated as a root-finding problem for a mismatch function $M(E)$. For a given trial energy E , the Schrödinger equation is integrated and a boundary mismatch is evaluated. Because the potential is symmetric, the half-domain solution is enough:

once parity is chosen, the left half of the wavefunction is fixed by reflection and the unknown energy is the only quantity left to determine. The code uses two related formulations. For the infinite square well, finite square well, and quartic double well, the half-domain problem is shot outward from $x = 0$ with parity-consistent startup values and the raw mismatch is the outer boundary value $M(E) = \psi_E(x_{\max})$. This outward formulation is the more general one in the project, because the startup comes from the exact parity conditions at the symmetry point and therefore applies to any symmetric potential once the half-domain reduction is valid. For the harmonic oscillator and the RK4 comparison, the code instead shoots inward from $x_{\max}$ through the decaying tail and enforces parity at the origin:

$$\text{even states: } M(E) = \psi'_E(0), \quad \text{odd states: } M(E) = \psi_E(0) \quad (2)$$

The inward formulation is more specialized: it assumes that the chosen truncation point lies in a forbidden tail where the physical solution is locally exponential, so it is used here for smooth confining infinite-domain problems rather than as a universal replacement for the outward method. The two formulations also differ slightly in how the brackets are polished once a sign change has been found. The outward solver bisection the wall-mismatch bracket and then performs a few safeguarded secant-style updates inside the final sign-changing interval, which improves the final raw boundary residual without giving up bracketing robustness. The inward and RK4 solvers use straight bisection on the origin-parity mismatch instead, which keeps that specialized formulation simpler and fully bracketed. Eigenvalues correspond to the roots of $M(E) = 0$, which are located by bracketing sign changes and then refined numerically. To make the procedure transparent, the report now uses two complementary diagnostics: a global raw-mismatch scan over the full energy interval, where only the final root estimates are marked, and per-root zoom plots that restrict the horizontal axis to one initial bracket and show the full bisection history. Zero crossings identify the allowed energies.

In the implementation, the inward even-state mismatch uses a fourth-order backward finite-difference stencil to approximate $\psi'(0)$ from the last few points of the inward integration grid. This is important because a lower-order boundary derivative would spoil the fourth-order behavior otherwise expected from the Numerov recurrence in the harmonic-oscillator study.

3 Results

Each potential is used to validate, test, or demonstrate different aspects of the numerical method, ranging from analytic benchmarks to nontrivial quantum behavior.

3.1 Infinite square well

The infinite square well serves as a benchmark problem with known analytic spectrum:

$$E_n = \frac{(n+1)^2 \pi^2}{8a^2} \quad (3)$$

The potential is:

$$V(x) = \begin{cases} 0 & , |x| \leq a \\ \infty & , |x| > a \end{cases} \quad (\text{implemented numerically with a large finite } V_0) \quad (4)$$

In the production calculation, the well half-width is $a = 1$, the outward-shooting half-domain uses $x_{\max} = a = 1$, the Numerov grid uses $n_{\text{grid}} = 2500$ points, and the numerical wall height is set to $V_0 = 10^6$. The grid-convergence study for this case uses $n_{\text{grid}} \in \{50, 80, 120, 180\}$ at fixed $x_{\max} = 1$.

The numerical eigenvalues agree with the analytic values to essentially machine precision, as shown in Fig. 1. In the saved run the first four relative errors are 4.9×10^{-11} , 1.3×10^{-10} ,

2.6×10^{-11} , and 7.0×10^{-12} . This is exactly the behavior expected for a benchmark problem of this type: the energies follow the analytic law $E_n \propto (n+1)^2$, and the numerical points are visually indistinguishable from the exact curve on the scale of the plot.

The probability densities for the first four states (Fig. 2) also have the expected structure. The ground-state density is maximal at the center, while the excited states show the correct sequence of interior nodes inherited from the alternating even/odd parity eigenfunctions. All densities vanish at $x = \pm a$, consistent with the hard-wall boundary condition.

The shooting method is validated by plotting the boundary mismatch functions for both even and odd sectors (Fig. 3). The even-sector scan has zero crossings near $E \approx 1.234$ and $E \approx 11.103$, while the odd-sector scan crosses zero near $E \approx 4.935$ and $E \approx 19.739$. These are precisely the first four physical eigenvalues, so the parity separation and bracketing logic are behaving as intended rather than picking spurious roots.

The convergence of the energy error as a function of grid spacing h (Fig. 5) is also as expected for Numerov. The fitted exponents from the saved convergence study are $p = 3.94, 4.00, 3.98$, and 4.00 for the first four states, which is strong evidence that the recurrence, startup values, and boundary treatment are all consistent with fourth-order accuracy. The higher states sit at larger absolute error for a fixed grid, which is also expected because their shorter wavelengths are harder to resolve.

Finally, the wavefunctions for the first four states are plotted together with their corresponding energies and the confining potential (Fig. 6). The alternating parity, increasing node count, and quadratic growth of the energy levels all match the analytic infinite-well picture.

3.2 Harmonic oscillator

3.2.1 Numerov method on unbounded domains

For the harmonic oscillator with $V(x) = \frac{1}{2}\omega^2 x^2$, the exact spectrum is:

$$E_n = \omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots \quad (5)$$

This provides a second analytic benchmark, but on an unbounded domain. Numerically, the domain is truncated to $[-x_{\max}, x_{\max}]$, and decaying boundary conditions are imposed. In the production Numerov run, the parameters are $\omega = 1$, $x_{\max} = 8$, and $n_{\text{grid}} = 2500$. The grid-convergence study uses $n_{\text{grid}} \in \{500, 800, 1200, 1800, 2500\}$ at fixed $x_{\max} = 8$, while the box-size study uses $x_{\max} \in \{4, 5, 6, 7, 8, 10\}$ at approximately fixed spacing.

The numerical eigenvalues agree with the exact linear spectrum to extremely high precision, as shown in Fig. 7. In the saved run the first four relative errors are 6.2×10^{-13} , 2.3×10^{-12} , 6.6×10^{-12} , and 7.5×10^{-12} . This is exactly what should happen for the harmonic oscillator benchmark: the inward-shooting formulation suppresses contamination by the exponentially growing forbidden-region solution and cleanly recovers the exact ladder $E_n = \omega(n + \frac{1}{2})$.

The probability densities (Fig. 8) show the expected structure: the ground state is centered and Gaussian-like, while the excited states alternate in parity and gain one additional node at each step. The support also broadens with increasing energy, consistent with the classical turning points moving outward. This confirms both physical correctness and stability of the integration on the truncated domain.

The shooting mismatch functions (Fig. 9) exhibit clear zero crossings at the expected energies. In the even sector the roots lie near $E \approx 0.5$ and $E \approx 2.5$, while in the odd sector they lie near

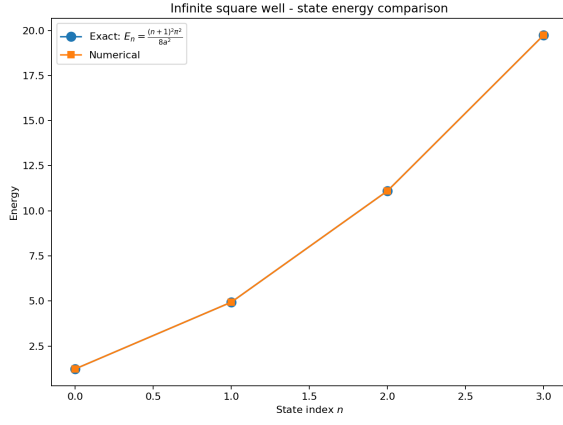


Figure 1: Comparison between numerical and exact energy eigenvalues for the infinite square well. The agreement is excellent for the first four states.

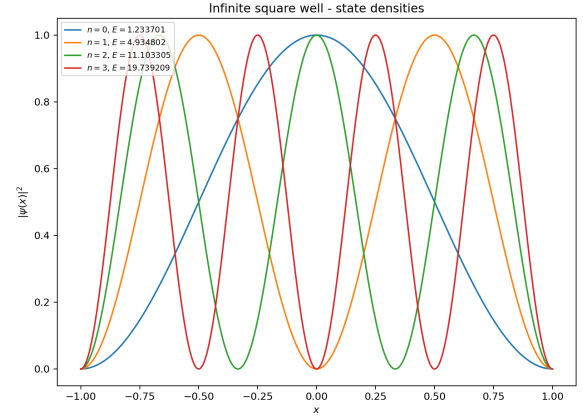
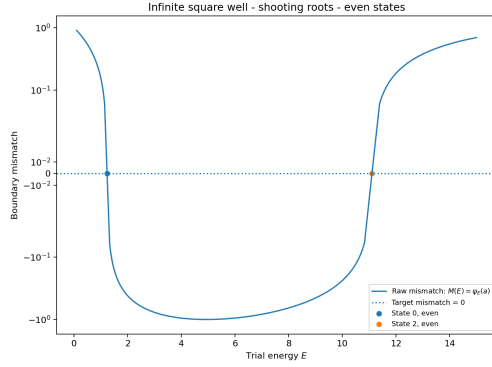
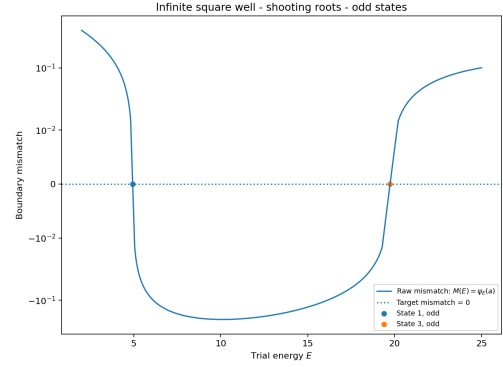


Figure 2: Probability densities $|\psi(x)|^2$ for the first four eigenstates of the infinite square well. The correct number of nodes and boundary behavior are clearly reproduced.

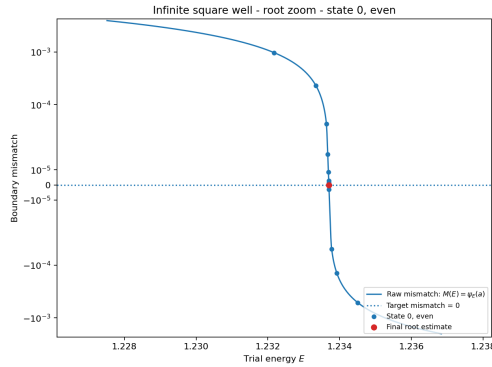


(a)

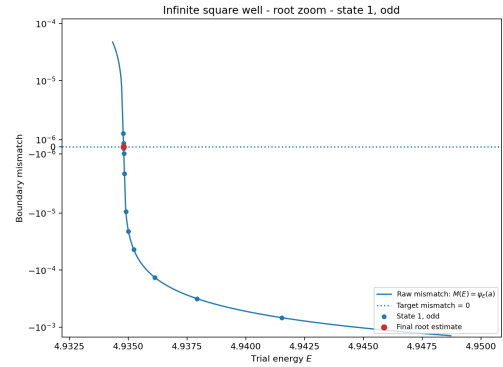


(b)

Figure 3: Raw boundary mismatch functions for even (a) and odd (b) states. The zero crossings correspond to the physical eigenvalues found by the shooting method.

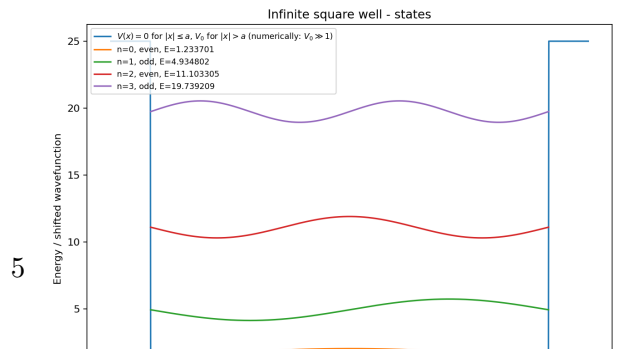
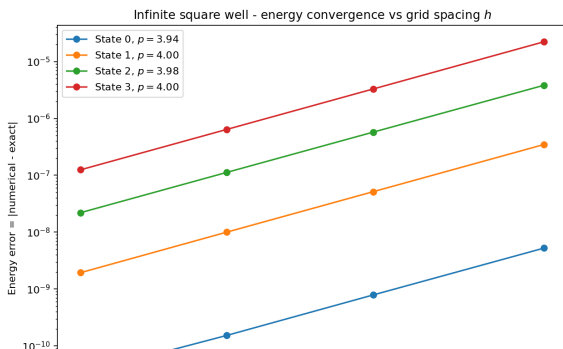


(a)



(b)

Figure 4: Zoomed root brackets for the lowest even (a) and odd (b) infinite-well states. These panels show the raw mismatch only inside one initial sign-changing bracket, together with the recorded bisection iterates.



$E \approx 1.5$ and $E \approx 3.5$. The raw mismatch spans many orders of magnitude because the inward integration starts in the decaying tail, but the sign changes remain sharp and the roots alternate even–odd as they should. The corresponding zoomed brackets in Fig. 10 make that local sign change explicit and show the final bisection convergence in the neighborhood of a single root rather than over the full dynamic range of the scan.

Convergence with respect to grid spacing h (Fig. 11) is overall consistent with fourth-order behavior. The saved study gives fitted slopes $p \approx 5.30, 3.97, 4.57$, and 4.00 for the first four states. The two slopes above four are not evidence of true superconvergence: they occur only after the errors have already dropped to the 10^{-11} – 10^{-12} range, where root-finding tolerance and floating-point effects distort a short log–log fit. The raw errors themselves still decrease in the expected high-order way, from about 3.2×10^{-9} – 2.6×10^{-8} on the coarsest grid to 5.1×10^{-13} – 2.6×10^{-11} on the finest grid.

The dependence on domain size in Fig. 12 shows the expected truncation-to-discretization crossover. At $x_{\max} = 4$ the tail is clipped too aggressively and the fourth-state error is still about 5.9×10^{-5} . By $x_{\max} = 5$ the errors have already fallen dramatically, and by $x_{\max} \approx 6$ the curves flatten near the discretization floor: roughly 10^{-12} for the ground state, a few 10^{-12} for the first odd state, and about 10^{-11} – 10^{-10} for the higher states. This confirms that the dominant error for large boxes is no longer domain truncation.

Finally, the combined state plot (Fig. 13) shows the wavefunctions together with the confining potential. The increasing spatial extent and alternating parity agree with the analytic harmonic-oscillator picture.

3.2.2 Numerov versus Runge–Kutta

As an additional method comparison, the harmonic oscillator was also solved using a fourth-order Runge–Kutta (RK4) shooting method. In this formulation the second-order Schrödinger equation is rewritten as the first-order system:

$$\psi' = \phi, \quad \phi' = 2[V(x) - E]\psi \quad (6)$$

This makes RK4 a useful general-purpose comparison for Numerov, which instead exploits the special second-order structure of the Schrödinger equation directly. The RK4 comparison uses the same physical parameters $\omega = 1$ and $x_{\max} = 8$ as the Numerov run, and it reuses the same family of grid spacings derived from the Numerov convergence study so the two methods are compared on matched meshes. The representative RK4 spectrum plot uses $n_{\text{grid}} = 1200$, and the RK4 box-size study again takes $x_{\max} \in \{4, 5, 6, 7, 8, 10\}$ at approximately fixed spacing.

Figure 14 compares the maximum energy error over the computed harmonic-oscillator states as a function of grid spacing. Both methods show approximately fourth-order convergence, as expected. The RK4 slopes are much more uniformly close to four than the Numerov fit, with saved values $p \approx 4.04, 4.02, 4.01$, and 4.00 . On the saved grids, the maximum absolute error ranges from 2.6×10^{-8} down to 2.6×10^{-11} for Numerov and from 6.0×10^{-8} down to 9.6×10^{-11} for RK4, so Numerov is consistently better by a factor of about 2–4 at fixed h even though both methods are fourth order.

The box-size study tells the same story for both methods. At fixed spacing, increasing $x_{\max}$ from 4 to 6 sharply reduces the truncation error, after which both methods level off at their respective discretization floors. This is a useful reminder that method order alone is not the whole story: domain truncation, root-finding tolerance, and boundary diagnostics all matter. After using a high-order boundary derivative for the Numerov inward-shooting condition, the

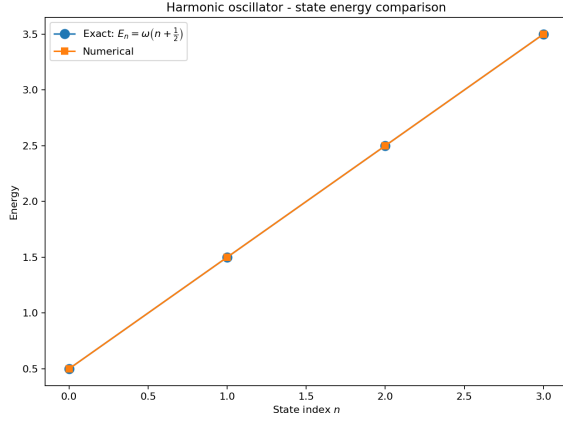


Figure 7: Comparison between numerical and exact energy eigenvalues for the harmonic oscillator. The agreement is excellent for the first four states.

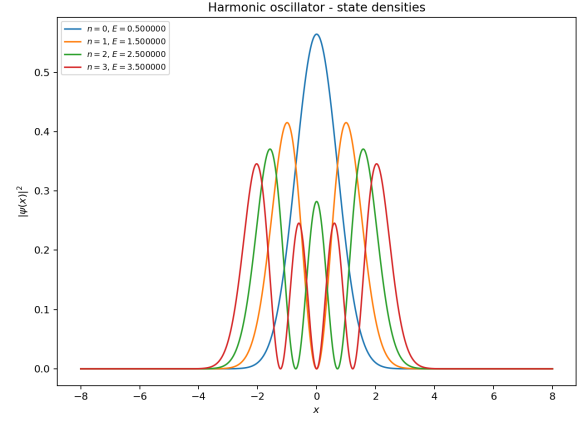
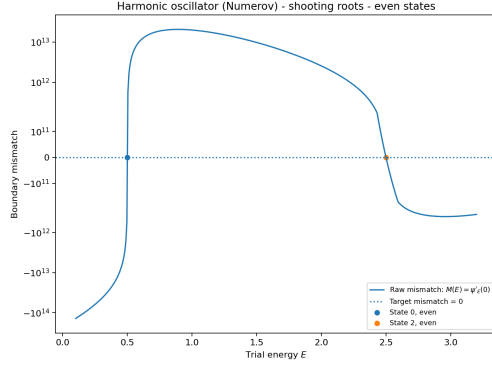
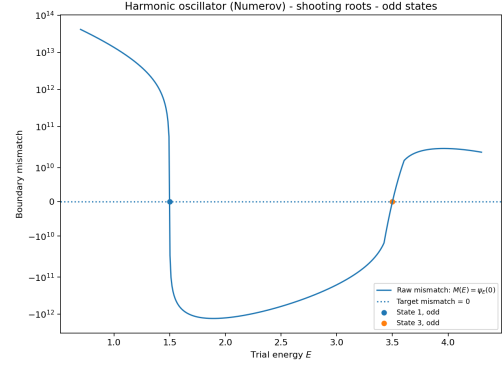


Figure 8: Probability densities $|\psi(x)|^2$ for the first four harmonic oscillator eigenstates. The correct node structure is clearly reproduced.

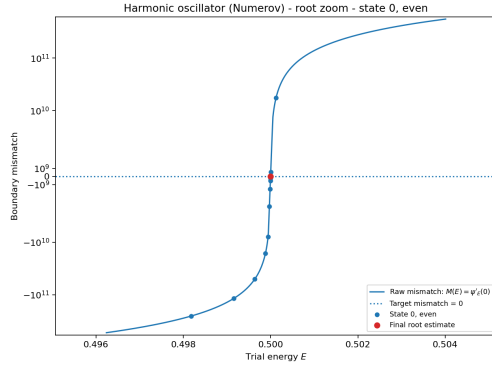


(a)

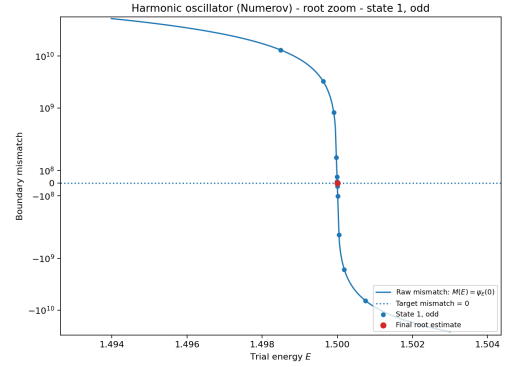


(b)

Figure 9: Raw boundary mismatch functions for even (a) and odd (b) states. The zero crossings correspond to the physical eigenvalues.



(a)



(b)

Figure 10: Zoomed root brackets for the lowest even (a) and odd (b) harmonic-oscillator states. These local views remove the global dynamic-range compression and show the raw mismatch together with the bisection iterates inside each initial bracket.

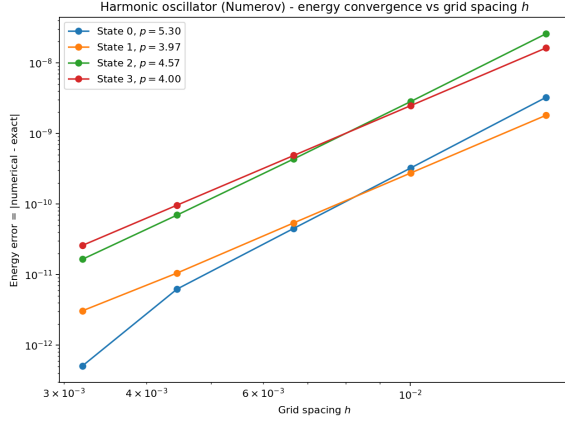


Figure 11: Convergence of the energy error as a function of grid spacing h . The slopes are close to fourth order, as expected for the Numerov method.

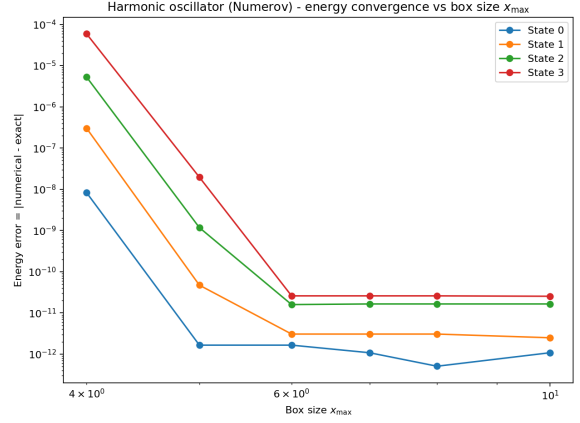


Figure 12: Dependence of the energy error on the domain size $x_{\max}$. Errors decrease rapidly before reaching a plateau set by discretization and floating-point limits.

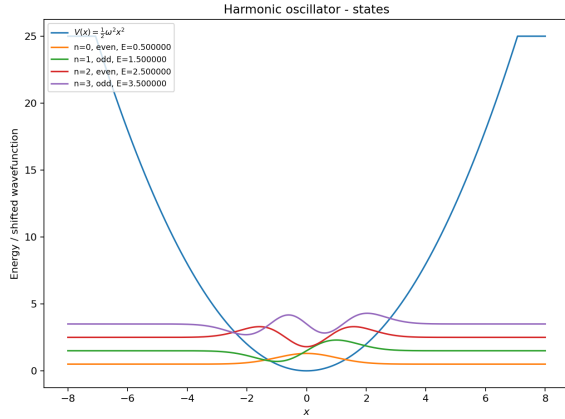


Figure 13: Wavefunctions for the first four harmonic oscillator states plotted together with the potential.

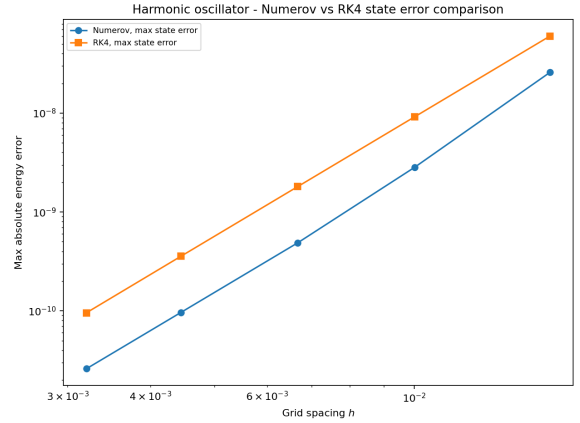


Figure 14: Comparison of Numerov and RK4 shooting for the harmonic oscillator. Both methods show fourth-order convergence, while Numerov gives a smaller maximum energy error at the same grid spacing.

Numerov and RK4 comparison behaves as expected and gives Numerov the smaller error at matched grid spacing.

3.3 Double-well potential

The quartic double-well potential is used as the main nontrivial bound-state example. In the shifted form used here:

$$V(x) = ax^4 - bx^2 - V_{\min}, \quad V_{\min} = -\frac{b^2}{4a} \quad (7)$$

the two minima are shifted to zero. This shift is applied with the analytic minimum rather than the sampled grid minimum $\min_i V(x_i)$ taken over the current mesh points, because the sampled minimum drifts slightly as the grid changes and would otherwise contaminate convergence studies. The analytic shift also gives a cleaner energy reference: the well bottoms sit at zero, so the reported low-lying energies can be read directly relative to the bottoms of the wells, while the potential shape and tunneling splittings remain unchanged. Unlike the infinite square well and harmonic oscillator, this system does not have a simple analytic spectrum, so the validation relies on internal consistency checks, convergence studies, parity structure, and physically expected tunneling behavior. The production run uses $a = 1$, $b = 6$, the analytic minimum shift $V_{\min} = b^2/(4a)$, a box size $x_{\max} = 4$, and $n_{\text{grid}} = 4000$. The successive-refinement study uses $n_{\text{grid}} \in \{600, 1000, 1600, 2200, 3000, 4000\}$, the plotted box-size study uses $x_{\max} \in \{2.5, 3.0, 3.5, 4.0, 4.5\}$ at approximately fixed spacing, and the box errors are measured against a separate larger-box reference run at $x_{\max} = 5.0$. The splitting sweep varies $b \in \{3, 4, 5, 6, 7, 8\}$ with $a = 1$.

Figure 15 shows the first four computed states together with the double-well potential. The two lowest states form a nearly degenerate even-odd pair, with energies $E_0 \approx 2.357273$ and $E_1 \approx 2.359372$, so the splitting is only about 2.10×10^{-3} . This is the expected tunneling doublet: the even and odd states are symmetric and antisymmetric combinations of states localized in the two wells. The next pair is much higher in energy, with $E_2 \approx 6.548814$ and $E_3 \approx 6.684430$, and its splitting is much larger because those states overlap more strongly through the central barrier.

The probability densities in Fig. 16 show the expected localization near the two wells. The lowest pair has nearly identical density, because the sign difference between even and odd wavefunctions disappears in $|\psi(x)|^2$. The higher states show additional node structure and broader support, as expected for excited double-well states. The figure also makes the tunneling interpretation concrete: the low-lying states are concentrated in the two minima near $x = \pm\sqrt{b/2}$, while the barrier region near the origin is sampled only through the small connecting tail.

The root-finding diagnostics (Fig. 17) show well-defined zero crossings in the even and odd parity sectors. For the saved run, the first even-sector roots occur near $E \approx 2.36$ and $E \approx 6.55$, while the first odd-sector roots occur near $E \approx 2.36$ and $E \approx 6.68$. The corresponding zoom views in Fig. 18 confirm that these are genuine local sign changes of the raw mismatch and not artifacts of the large boundary amplitudes seen in the full scan.

Since there is no closed-form exact spectrum for this potential, grid convergence is measured using successive refinements at fixed domain size: the plotted surrogate error is the difference $|E(h_i) - E(h_{i+1})|$ between each grid and the next finer one, rather than the difference to one fixed finite reference run. The convergence with respect to grid spacing (Fig. 19) still shows stable high-order behavior, with fitted slopes $p \approx 4.18, 3.95, 4.55$, and 5.67 for the first four states. The values above four should not be overinterpreted as true superconvergence. They appear only when the successive-difference errors have already fallen to about 10^{-12} – 10^{-13} for the low states and 10^{-11} for the higher states, so short log–log fits are being distorted by the numerical

floor. What matters physically is that every state improves systematically under refinement and the dominant trend remains high-order decay of the error.

The box-size study (Fig. 20) shows an especially important numerical effect for this problem. At $x_{\max} = 2.5$ the box is clearly too small: the errors are still of order 10^{-1} . At $x_{\max} = 3.0$ they have dropped to the 10^{-4} – 10^{-3} range, and by $x_{\max} = 3.5$ they are already down near 10^{-8} . Once the box reaches $x_{\max} \approx 4.0$, the remaining error is only about 10^{-12} relative to the larger-box reference, so the final reported states are no longer dominated by artificial boundary truncation. This is exactly why the production calculation uses a larger box than the earlier, visibly contaminated small-box runs.

Finally, Fig. 21 shows how the lowest even-odd splitting changes with the double-well parameter b . As b increases from 3 to 8, the splitting falls from about 3.52×10^{-1} to 1.59×10^{-5} . This is the expected physical signature of tunneling suppression: increasing b deepens and separates the wells, raises the effective barrier between them, and therefore reduces the overlap between the left- and right-localized components.

Overall, the double-well results demonstrate that the solver works beyond analytic benchmarks. The computed states have the expected parity, localization, and near-degenerate tunneling structure; the splitting with respect to b behaves physically; and the convergence studies show that the numerical results are stable with respect to both grid spacing and domain size once the computational box is chosen large enough.

3.4 Finite square well

The finite square well is included as an additional nontrivial bound-state test. Unlike the infinite square well, the potential walls are finite, so bound states can leak into the classically forbidden region. The potential used here has the form:

$$V(x) = \begin{cases} 0 & , |x| \leq a \\ V_0 & , |x| > a \end{cases} \quad (8)$$

and only states with $E < V_0$ are true bound states. In the production run, the well parameters are $a = 1$ and $V_0 = 12$, while the outward-shooting box uses $x_{\max} = 4$ and the Numerov mesh uses $n_{\text{grid}} = 3000$. A supplementary grid-refinement check recomputes the first four bound states on $n_{\text{grid}} \in \{400, 700, 1000, 1500, 2200\}$ and compares those energies to the finest saved run, so the plotted quantity is $|E_N - E_{\text{ref}}|$ with the $n_{\text{grid}} = 3000$ spectrum acting as the practical reference. Unlike the smooth benchmark potentials, however, the finite square well is only piecewise smooth: $V(x)$ jumps at $|x| = a$. That discontinuity degrades the observed asymptotic convergence order, so this refinement study should be read mainly as a practical convergence check rather than a clean fourth-order Numerov benchmark.

Figure 22 shows the first four computed states together with the finite well potential. For the parameter choice $V_0 = 12$ and $a = 1$, the four reported bound-state energies are $E_0 \approx 0.847846$, $E_1 \approx 3.344466$, $E_2 \approx 7.294436$, and $E_3 \approx 11.782210$. The first three lie comfortably below the barrier and are localized mainly inside the well, with exponentially decaying tails outside. The fourth state is only about 0.218 below the barrier top, so it is a near-threshold bound state and therefore shows much stronger leakage into the exterior region. This is exactly the behavior expected for a finite well, where weaker binding implies a longer evanescent decay length outside the classically allowed region.

The probability densities in Fig. 23 show the same effect even more clearly. The low-lying states are concentrated inside the well and retain the expected even-odd node ordering, while the highest plotted state has substantial weight outside $|x| \leq a$. That long exterior tail is physically

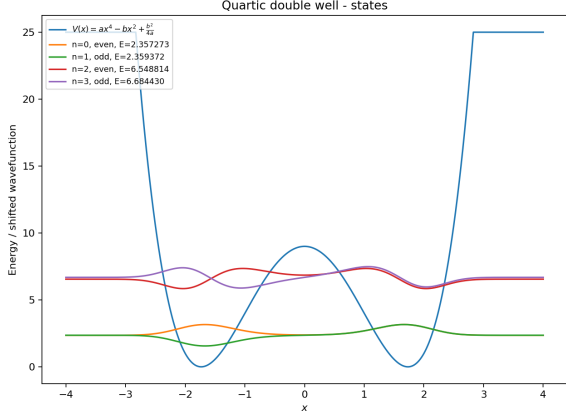


Figure 15: First four bound states of the quartic double-well potential. The lowest two states form a nearly degenerate tunneling doublet.

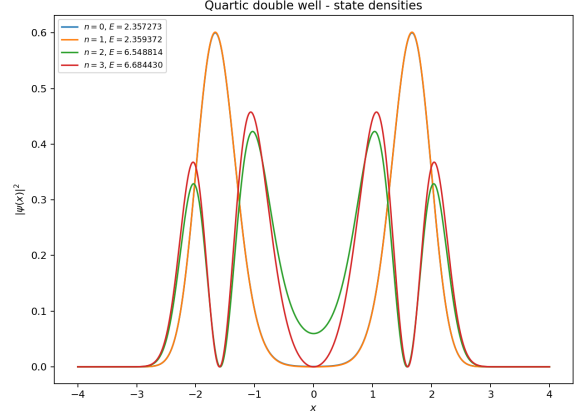
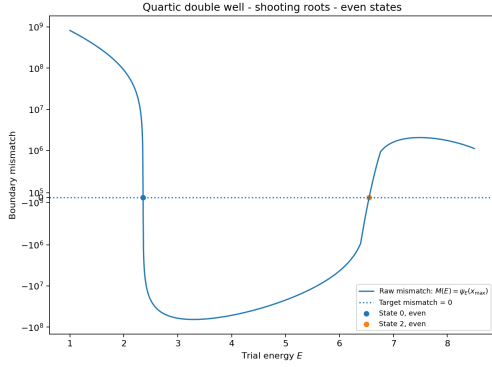
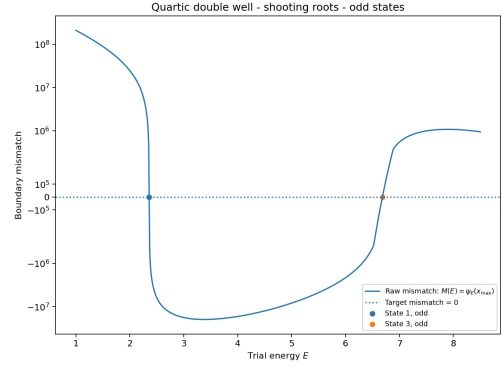


Figure 16: Probability densities $|\psi(x)|^2$ for the first four double-well states. The lowest even and odd states have nearly identical densities, as expected for a tunneling doublet.

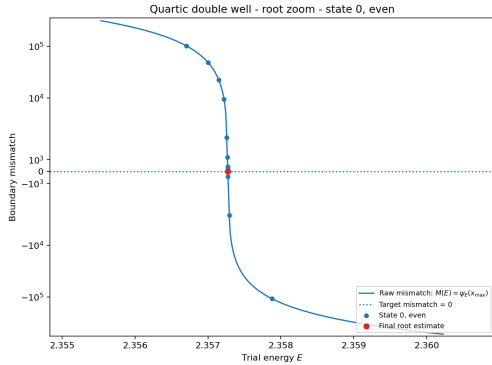


(a)

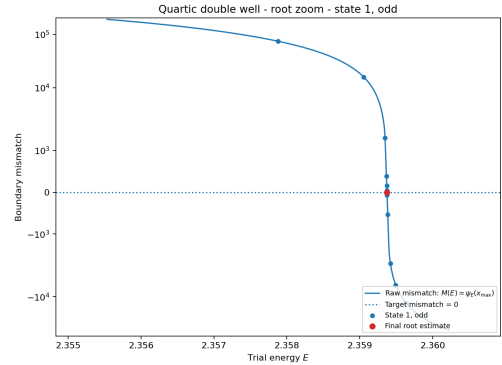


(b)

Figure 17: Raw shooting mismatch functions for even (top) and odd (bottom) double-well states. The zero crossings define the numerical eigenvalues.



(a)



(b)

Figure 18: Zoomed root brackets for the lowest even (a) and odd (b) double-well states. These local panels show the raw mismatch and the recorded bisection sequence inside the initial sign-changing brackets.

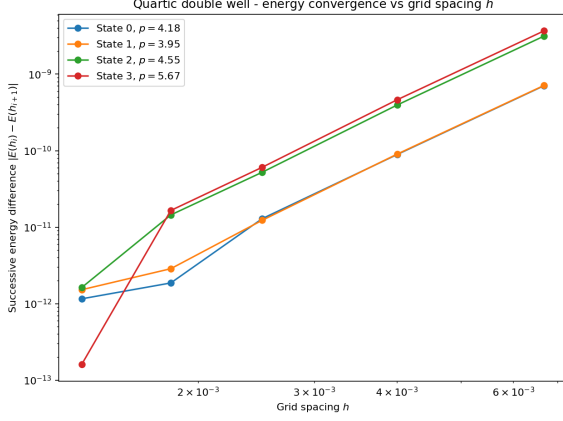


Figure 19: Grid convergence for the quartic double well. Since no exact spectrum is available, the vertical axis shows the successive-refinement surrogate $|E(h_i) - E(h_{i+1})|$ rather than $|E_{\text{numerical}} - E_{\text{exact}}|$. The curves still decrease with clear high-order behavior under grid refinement.

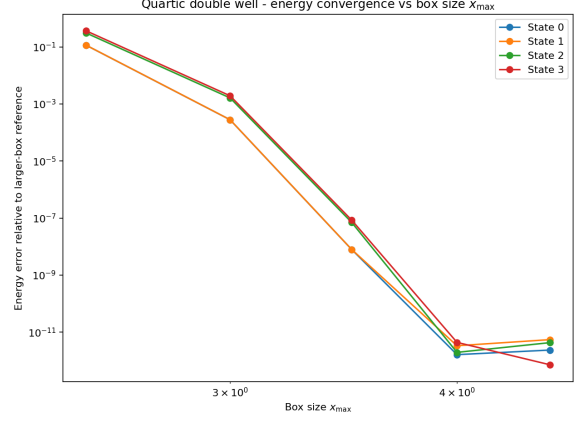


Figure 20: Dependence of the double-well energy errors on box size, measured relative to a separate larger-box reference calculation at approximately fixed grid spacing. The errors decrease rapidly as the domain is enlarged and then reach a numerical plateau.

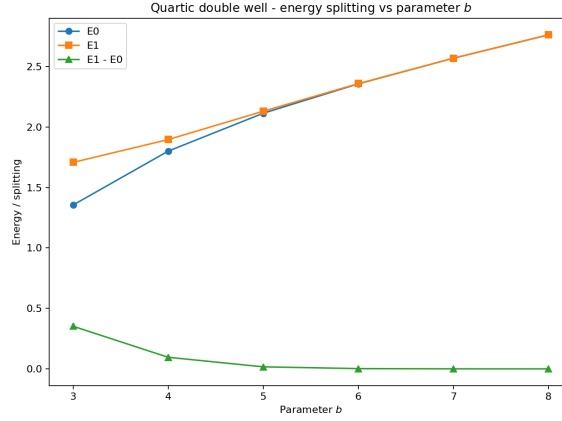


Figure 21: Lowest two double-well energies together with their splitting as the parameter b is varied. The splitting decreases as the barrier becomes stronger, consistent with tunneling suppression.

correct: unlike the infinite well, a finite barrier does not force the wavefunction to vanish at the walls, it only makes the outside amplitude decay exponentially.

The finite-well root diagnostics in Fig. 24 add an important numerical interpretation. In the even-sector global scan, the raw mismatch becomes small again as E approaches the barrier height $V_0 = 12$ from below. This does not indicate an additional bound state. As $E \rightarrow V_0^-$, the exterior decay constant $\kappa = \sqrt{2(V_0 - E)}$ tends to zero, so on a fixed box the boundary value $\psi_E(x_{\max})$ can decrease in magnitude without actually crossing zero. The zoomed near-threshold brackets in Fig. 25 show the correct interpretation: the reported high-energy roots are genuine sign changes, but the later drift of the raw mismatch toward zero remains one-sided and therefore does not create more bound states. Since true bound states require $E < V_0$, extending the scan to $E > V_0$ would enter continuum behavior rather than reveal extra discrete levels.

The finite-well grid-refinement study provides a modest quantitative check in addition to these qualitative diagnostics: the first few bound-state energies converge toward the finest-grid reference as h decreases, but the observed slope is noticeably lower than in the smooth benchmark cases because the discontinuous well edge dominates the error. Even so, this part of the project remains primarily an additional physical test rather than the main convergence showcase. In particular, the finite-well figures confirm that the solver distinguishes strongly bound interior states from a near-threshold state with pronounced barrier penetration.

3.5 Scattering and resonant tunneling

3.5.1 Single barrier potential

As a secondary extension beyond bound states, the Numerov framework is also applied to one-dimensional scattering from a finite rectangular barrier. For each incident energy E , the complex wavefunction is propagated through the barrier region and decomposed into incident, reflected, and transmitted waves. This gives the transmission and reflection probabilities $T(E)$ and $R(E)$. The integration is started from the right free region, not the left, because the right boundary condition is the simplest one to impose directly: for a left-incident scattering state, the far-right asymptotic solution contains only the transmitted outgoing wave e^{ikx} . The left asymptotic region is more complicated because it contains a superposition of incident and reflected waves, so those amplitudes are recovered afterward by decomposing the numerically propagated left solution.

The shared scattering grid for both barrier problems is $x \in [-8, 8]$ with $n_{\text{grid}} = 4000$ points. For the single-barrier run, the barrier parameters are $V_0 = 5$, width 1.2, and center 0, while the incident energy is sampled on 240 points from $E = 0.2$ to $E = 10$.

The single-barrier result is shown in Fig. 26. For the saved run the barrier has height $V_0 = 5$ and width 1.2, while the incident energy is scanned from 0.2 to 10. At low incident energy, where E is well below the barrier height, transmission is strongly suppressed and reflection is close to one; for example, at $E = 0.2$ the calculation gives $T \approx 3.6 \times 10^{-4}$ and $R \approx 0.99964$. As the incident energy increases through and above the barrier, transmission rises and reflection falls, which is the expected qualitative behavior for a finite barrier.

The curve also shows an important physical detail: even for $E > V_0$, transmission is not exactly one. The barrier still causes partial reflection because the wave number changes across the interfaces. In the saved data, $T(E)$ reaches its maximum value very close to one near $E \approx 8.4$, but by $E = 10$ it is still only about 0.956, with a residual reflection of about 0.044. That is fully consistent with above-barrier scattering from a finite rectangular barrier rather than a sharp classical step from total reflection to total transmission.

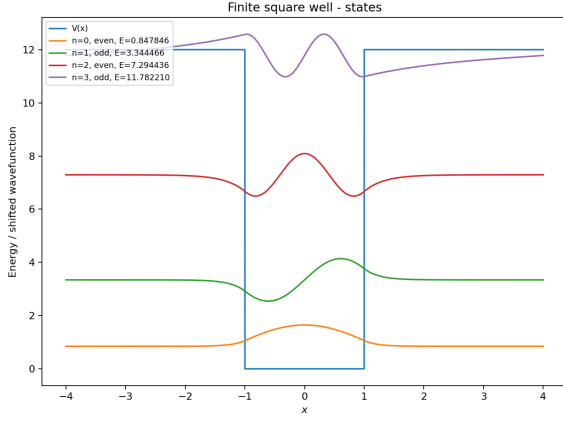


Figure 22: First four finite square well states plotted together with the finite barrier potential. States below the barrier are localized in the well, while the near-threshold state has a larger evanescent tail.

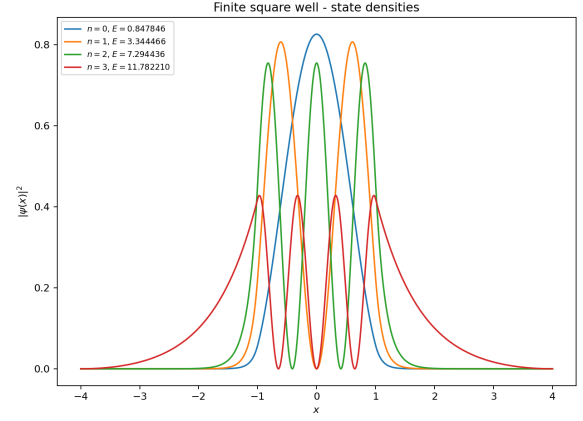
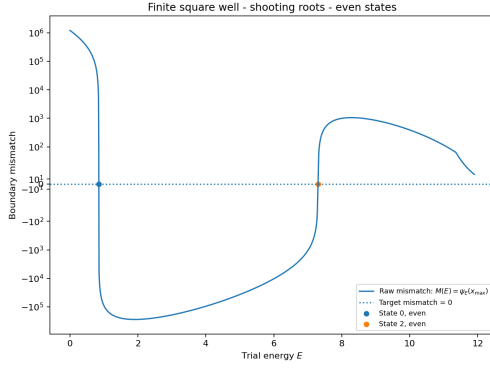
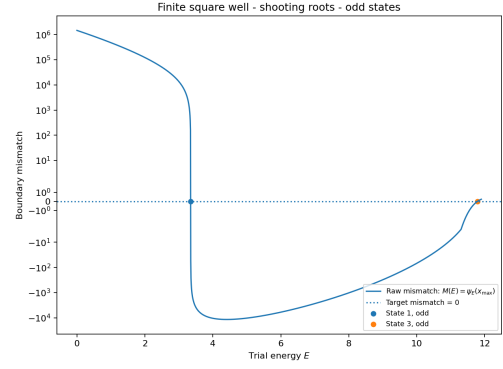


Figure 23: Probability densities $|\psi(x)|^2$ for the first four finite square well states. The densities show both interior oscillation and exterior exponential leakage.

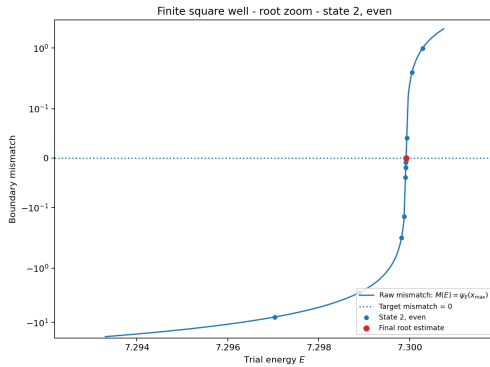


(a)

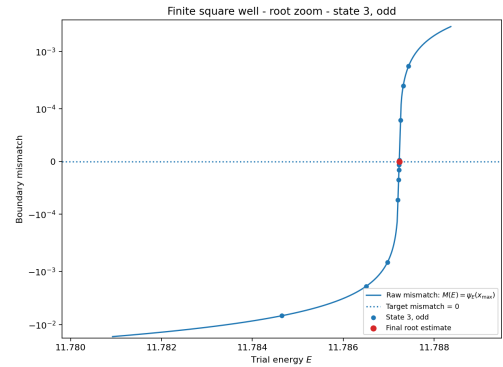


(b)

Figure 24: Raw shooting mismatch functions for even (a) and odd (b) finite-well states. The global scans include the near-threshold regime just below the barrier top V_0 .



(a)



(b)

Figure 25: Zoomed near-threshold brackets for the higher even (a) and odd (b) finite-well bound states. These views isolate the genuine sign-changing roots close to the barrier top and show the full recorded bisection history.

The dashed curve shows the conservation check $T(E) + R(E)$, which is numerically indistinguishable from one across the full energy range in the saved data: the largest deviation is about 2.3×10^{-12} . This confirms that the numerical scattering calculation conserves probability flux to within floating-point error for this setup.

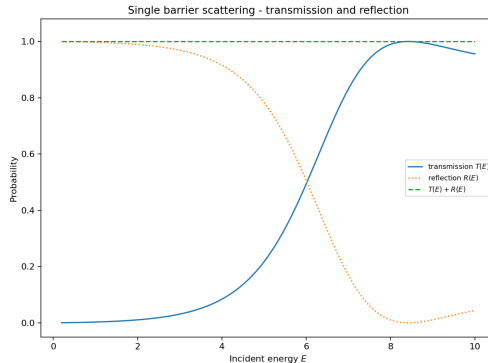


Figure 26: Transmission and reflection probabilities for scattering from a finite rectangular barrier. At low energies reflection dominates, while at high energies transmission approaches unity. The conservation check $T(E) + R(E)$ remains close to one.

3.5.2 Double barrier potential

The double-barrier potential is used to demonstrate resonant tunneling. In this case the two barriers create an intermediate well, so certain incident energies can form quasi-bound states between the barriers. At these energies, destructive reflection and constructive transmission lead to sharp peaks in the transmission coefficient.

For the production double-barrier run, each barrier has height $V_0 = 5$, the barrier width is 0.6, the central well width is 1.4, the structure is centered at $x = 0$, and the incident energy is sampled on 420 points from $E = 0.2$ to $E = 5.0$. Resonance candidates are then identified from the transmission curve using a simple peak threshold of $T(E) > 0.65$.

Figure 27 shows the computed transmission and reflection probabilities for two barriers separated by a central well. The saved transmission curve contains two clear resonances: a very narrow low-energy peak at $E \approx 1.13938$ with $T \approx 0.99996$, and a broader high-energy peak at $E \approx 4.35847$ with $T \approx 0.99987$. At both resonant energies, reflection is almost completely suppressed. This is the expected behavior for resonant tunneling: when the incident energy matches a quasi-bound mode of the region between the barriers, multiple reflections inside the central well interfere constructively for transmission and destructively for reflection.

The conservation check $T(E) + R(E)$ is again essentially exact in the saved calculation, with a largest deviation of about 8.4×10^{-12} . This is an important diagnostic because it verifies that the numerical decomposition into incident, reflected, and transmitted components is self-consistent over the full energy scan, including very sharp resonances.

The two resonances also have visibly different widths, which is physically meaningful. The first peak is extremely narrow, indicating a longer-lived quasi-bound state with weaker leakage through the barriers, while the second peak is much broader, indicating stronger coupling to the external scattering channels. That broader width is why the transmission stays elevated over a larger energy interval around the second resonance.

Figure 28 shows the probability density at the first resonance. The wavefunction is strongly enhanced in the region between the two barriers, with only much smaller amplitude outside. This is the expected signature of a quasi-bound state trapped temporarily in the central well, and it explains the near-unit transmission at the corresponding resonant energy.

These results are sufficient for the double-barrier part of the report because the scattering section is a secondary extension. The plots demonstrate the key qualitative physics: transmission resonances, reflection suppression at resonance, flux conservation, and resonant enhancement of the wavefunction in the central well.

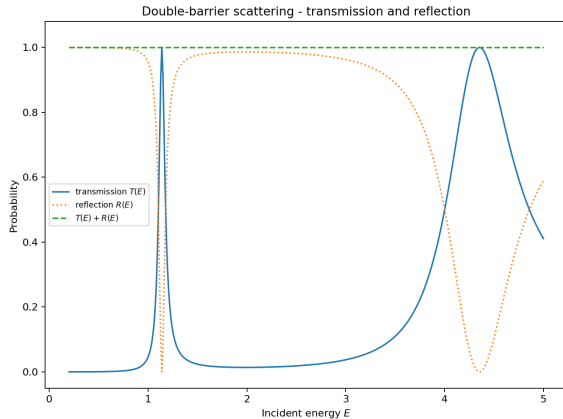


Figure 27: Transmission and reflection probabilities for the double-barrier potential. Sharp transmission peaks occur at resonant energies, where reflection is suppressed and $T(E) + R(E)$ remains close to one.

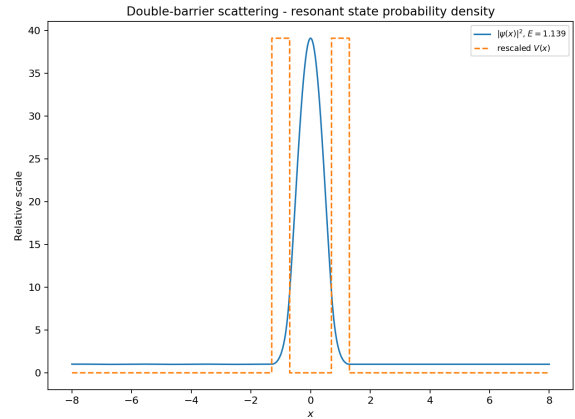


Figure 28: Probability density at the first resonant energy. The wavefunction is strongly enhanced between the two barriers, showing the quasi-bound state responsible for resonant tunneling.

4 Discussion

The calculation was checked in four complementary ways. First, an automated validation suite tests normalization, the fourth-order boundary derivative, analytic benchmark energies, the double-well level ordering, scattering flux conservation, and the entry-script import path. Second, the infinite square well and harmonic oscillator reproduce known analytic spectra, providing direct validation of the eigenvalue search. In the saved runs, the first four square-well relative errors are of order 10^{-11} – 10^{-10} , and the first four harmonic-oscillator relative errors are of order 10^{-12} – 10^{-11} . Third, the wavefunctions have the expected parity, node structure, normalization, and boundary behavior. Fourth, convergence tests with respect to grid spacing and domain size show that the results are stable under numerical refinement.

The convergence behavior also exposes important numerical details. In the infinite square well, the observed fourth-order convergence confirms that the Numerov recurrence, startup conditions, and boundary treatment are consistent. For the harmonic oscillator, the box-size study separates domain truncation error from discretization error: increasing $x_{\max}$ initially reduces the error rapidly, after which the error reaches a plateau set by grid spacing, root-finding tolerance, and floating-point limits. Slopes slightly above fourth order in some fits are therefore interpreted as small-error fitting artifacts rather than evidence of a truly higher-order method. The same caution applies in the double-well case, where successive-refinement fits above fourth order appear only once the differences are already near the numerical floor.

The comparison with RK4 shows why method order alone is not enough. RK4 evolves both ψ and ψ' directly, while Numerov requires numerical boundary diagnostics. Once the boundary derivative in the Numerov inward-shooting calculation is evaluated with comparable accuracy,

Numerov behaves as expected and gives smaller errors than RK4 at the same grid spacing for the harmonic oscillator, typically by a factor of about 2–4 in the saved max-error comparison.

For the quartic double well, where no closed-form spectrum is available, correctness is established through self-consistency rather than analytic comparison. The use of an analytic potential shift, separated grid and box convergence studies, parity-specific root diagnostics, and normalized boundary leakage all support the reliability of the computed states. The low-lying even-odd pair at $b = 6$ has the expected small tunneling splitting, about 2.10×10^{-3} , and the splitting decreases systematically to about 1.59×10^{-5} by $b = 8$, which is the expected physical signature of tunneling suppression.

The finite square well provides an intermediate physical check between analytic bound-state benchmarks and more complex tunneling problems. Its low-lying states remain localized inside the well, while the highest bound state lies only about 0.218 below the barrier and therefore shows the expected long evanescent tail outside the well. Its grid-refinement plot also illustrates an important numerical limitation: because the potential jumps at the well edge, the observed convergence order is lower than in the smooth benchmark cases even though the energies still converge correctly. This confirms that the same shooting framework handles both hard-wall-like confinement and finite barrier penetration.

The scattering examples are treated as a secondary extension. The single-barrier calculation shows the expected transition from reflection-dominated behavior at low energy to transmission-dominated behavior at high energy, while still retaining nonzero above-barrier reflection from interface mismatch. The double-barrier calculation shows two resonant transmission peaks, near $E \approx 1.13938$ and $E \approx 4.35847$, together with strong probability-density enhancement in the central well at resonance. In both cases, the saved calculations satisfy $T(E) + R(E) = 1$ to within about 10^{-11} – 10^{-12} , providing an additional numerical consistency check.

5 Conclusion

This project implemented a Numerov shooting solver for the one-dimensional time-independent Schrödinger equation. The method was validated on analytic bound-state problems, extended to finite and double-well potentials, and checked through convergence studies, parity diagnostics, and comparison with RK4.

The benchmark results are quantitatively strong. The infinite square well and harmonic oscillator reproduce the expected spectra to very high precision, with fourth-order convergence recovered once boundary conditions and startup terms are treated consistently. The harmonic-oscillator comparison also shows that the specialized Numerov scheme achieves smaller errors than RK4 at the same grid spacing when both are implemented carefully.

Beyond the analytic tests, the finite square well and quartic double well show the expected physical behavior: barrier penetration in the finite well, a near-degenerate tunneling doublet in the double well, and a splitting that decreases rapidly as the wells are deepened and separated. The scattering extension further reproduces the expected transition from ordinary tunneling through a single barrier to resonant transmission through a double barrier, with probability flux conserved to near machine precision throughout.

Overall, the results support the main conclusion of the project: Numerov shooting is an accurate and flexible method for one-dimensional quantum bound-state and scattering calculations, provided that boundary diagnostics, box size, and convergence checks are treated as part of the numerical method rather than as afterthoughts.

References

- [1] Tao Pang. *An Introduction to Computational Physics*. 2nd ed. Cambridge University Press, 2006.